Assignment-3: Naive Bayes Classifier

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***Abstract*—Income classification is the process of collecting and analyzing the data collected using census surveys to determine the beneficiaries of different public policies. Most of the classification approaches focus on feature selection or reduction. Some features are irrelevant and redundant, resulting in a lengthy detection process and degrades the classifier's performance. The purpose of this study is to identify essential input features in building an effective income classifier. In this paper, we aim to discover the relationship between different features and an individual's income. We apply one of the efficient classifier naive Bayes on datasets to determine where a person earn more than 50k annually or not.**

***Index Terms*—Classification, Bayes theorem, conditional probability, Gaussian distribution, features, predictors, classifier.**

I. Introduction

Classification of objects with many features and classes becomes difficult because estimating the probabilities would necessitate an enormous number of observations. The effect of a variable value on a given class is assumed to be independent of the values of other variables by Naive Bayes classifiers. This is known as class conditional independence. It is designed to simplify computation and is therefore considered naive. The assumption is relatively broad and is not always applicable. However, studies comparing classification algorithms have found that the naive Bayesian classifier performs similarly to classification trees and neural network classifiers. When applied to large databases, they have also demonstrated high accuracy and speed.

Naive Bayes Classifier is a classification technique where the target variable is categorical in nature. Abstractly, it is a **conditional probability model**. Given a data instance to be classified, represented by a vector $X_i = (x_{i1},\ldots,x_{in})$ representing n independent features, it assigns to this instance a probability $p(C_k|X_i)$ for each of K possible classes C_k. The probability $p(C_k|X_i)$ is calculated using the Bayes' Theorem which is stated mathematically as:

$$p(C_k|X_i) = \frac{p(C_k)p(X_i|C_k)}{p(X_i)}$$

where,
$p(C_k|X_i)$: Posterior probability
$p(C_k)$: Prior probability
$p(X_i|C_k)$: Likelihood
$p(X_i)$: Evidence
The data instance is then assigned with a class where the probability $p(C_k|X_i)$ is highest.

This paper looks at a conditional probability model that works on data extracted from the 1994 Census bureau database by Ronny Kohavi and Barry Becker (Data Mining and Visualization, Silicon Graphics). The key focus in this paper is to analyze the impact of various factors on an individuals income, identifying the most relevant features and using the data to create a naive Bayes classifier which will be able to identify the individuals with more than \$50k annual income. Ultimately, we would like to understand the data characteristics which affect the performance of naive Bayes classifier.

This paper is structured as follows. In the next section, we provide the necessary background and definitions. Section II focuses on the data description, analysis, handling of missing values and visualization to gain more insights into the problem, while Section III demonstrates that the use of *Naive Bayes Classifier*. A summary and conclusions are given in Section IV.

II. Mathematics behind Naive Bayes Classifier

This section focuses on the mathematical aspect of how to classify vectors of discrete-valued features, $x \in 1,\ldots,K^D$, where K is the number of values for each feature, and D is the number of features. A generative approach is employed. It requires us to specify the conditional class distribution, $p(x|y=c)$. The most straightforward approach is to assume the features are conditionally independent given the class label. Therefore the conditional class density as a product of one-dimensional densities can be written as:

$$p(x|y=c,\theta) = \prod_{j=1}^{D} p(x_j|y=c,\theta_{jc}) \tag{1}$$

The resulting model is called a **naive Bayes classifier (NBC)**. The class-conditional density depends on the type of each feature. Some probability distributions are described below:

- For real valued features Gaussian Distribution is used i.e. $p(x|y=c,\theta) = \prod_{j=1}^{D} \mathcal{N}(x_j|\mu_{jc},\sigma_{jc}^2)$ where μ_{jc} is the mean of feature j in objects of class c, and σ_{jc}^2 is its variance.
- A Bernoulli distribution is employed in case of binary features i.e. $p(x|y=c,\theta) = \prod_{j=1}^{D} \mathrm{Ber}(x_j|\mu_{jc})$ where μ_{jc} is the probability that feature j occurs in class c. This is known as **multivariate Bernoulli naive Bayes** model.

- For categorical features, $x_j \in 1, \ldots, K$, we can model using the multinoulli distribution: $p(x|y = c, \theta) = \prod_{j=1}^{D} \mathrm{Cat}(x_j|\mu_{jc})$, where μ_{jc} is a histogram over the K possible values for x_j in class c.

A. MLE for NBC

Probability for a single data is written as

$$p(x_i, y_i|\theta) = p(y_i|\pi) \prod_j p(x_{ij}|\theta_j) = \prod_c \theta_c^{\mathbb{I}(y_i=c)}$$

$$\prod_c \theta_c^{\mathbb{I}(y_i=c)} = \prod_c \prod_j p(x_{ij}|\theta_{jc})^{\mathbb{I}(y_i=c)} \tag{2}$$

Therefore, log-likelihood can be written as

$$\log p(\mathcal{D}|\theta) = \sum_{c=1}^{C} N_c \log \pi_c + \sum_{j=1}^{D} \sum_{c=1}^{C} \sum_{i:y_i=c} \log p(x_{ij}|\theta_{jc}) \tag{3}$$

(3) shows MLE getting decomposed into two terms, one concerning π and other having $D \times C$ terms containing θ_{jc}'s. Hence both terms can be optimized separately. The MLE for the class prior is given by the

$$\hat{\pi}_c = \frac{N_c}{N}; N_c = \sum_i \mathbb{I}(y_i = c) \tag{4}$$

The MLE for the likelihood terms depends on the assumption of type of distribution of each feature. In case of binary assumption ($x_j|y = c \sim \mathrm{Ber}(\theta_{jc})$) the MLE can is obtained as

$$\hat{\theta}_{jc} = \frac{N_{jc}}{N_c} \tag{5}$$

Models of this form are much more manageable, since they use prior probabilities of classes and independent probability distributions. If there are k classes and if a model can be expressed in terms of r parameters, then the corresponding naive Bayes model has $(k-1) + nrk$ parameters. In practice, often $k = 2$ (binary classification) and $r = 1$ (Bernoulli variables as features) are common, and so the total number of parameters of the naive Bayes model is $2n + 1$, where n is the number of binary features used for prediction.

B. Bayesian naive Bayes

The problem with the approach mentioned above is that it can overfit. For example, if an email spam classifier is constructed and word *subject* occurs in both classes, then MLE will estimate $\theta_{jc} = 1$ for it. What will happen if a new email is encountered which does not have this word in it? Algorithm will crash and burn, since $p(y = c|x, \hat{\theta})$ = 0 for both classes.

A simple solution is to use the Bayesian approach. For simplicity, a factored prior is assumed.

$$p(\theta) = p(\pi) \prod_{j=1}^{D} \prod_{c=1}^{C} p(\theta_{jc}) \tag{6}$$

A $\mathrm{Dir}(\alpha)$ prior for π and a $\mathrm{Beta}(\beta_0, \beta_1)$ prior for each θ_{jc} is used. Generally $\alpha = 1$ and $\beta = 1$ are used, corresponding to add-one or Laplace smoothing. [2]

Combining the factored likelihood in (3) with the factored prior above gives the following factored posterior:

$$p(\theta|\mathcal{D}) = p(\pi|\mathcal{D}) \prod_{j=1}^{D} \prod_{c=1}^{C} p(\theta_{jc}|\mathcal{D}) \tag{7}$$

$$p(\pi|\mathcal{D}) = \mathrm{Dir}(N_1 + \alpha_1, \ldots, N_C + \alpha_C) \tag{8}$$

$$p(\theta_{jc}|\mathcal{D}) = \mathrm{Beta}((N_c - N_{jc}) + \beta_0, N_{jc} + \beta_1) \tag{9}$$

In other words, to compute the posterior, the prior counts are updated with the empirical counts from the likelihood.

The training set can compute all model parameters (such as class priors and feature probability distributions). These are the probabilities' maximum likelihood estimates. The frequency-based probability estimate will be zero if a given class and feature value never occur together in the training set. When some probabilities are multiplied by this probability, they become zero. As a result, all probability estimates must include a small-sample correction to ensure that no probability is ever set to be precisely zero. The feature probability distribution will be calculated by comparing the range of values of a feature in a class to the range of values across all classes.

C. Using Model for Prediction

At test time, the goal is to compute

$$p(y = c|x, \mathcal{D}) \propto p(y = c|\mathcal{D}) \prod_{j=1}^{D} p(x_j|y = c, \mathcal{D}) \tag{10}$$

The correct Bayesian procedure is to integrate out the unknown parameters. Fortunately, this is easy to do, at least if the posterior is Dirichlet. In particular, the posterior predictive density can be obtained as $\frac{\alpha_j + N_j}{\alpha_0 + N}$. By simply plugging in the posterior mean parameters $\bar{\theta}$. Hence

$$p(y = c|x, \mathcal{D} \propto \bar{\pi}_c \prod_{j=1}^{D} (\bar{\theta}_{jc})^{\mathbb{I}(x_j=1)} (1 - \bar{\theta}_{jc})^{\mathbb{I}(x_j=0)} \tag{11}$$

$$\bar{\theta}_{jc} = \frac{N_{jc} + \beta_1}{N_c + \beta_0 + \beta_1} \tag{12}$$

$$\bar{\pi}_c = \frac{N_c + \alpha_c}{N + \alpha_0} \tag{13}$$

where $\alpha_0 = \sum_c \alpha_c$.

If tha psoterior is approximated by a single point, $p(\theta|\mathcal{D}) \approx \delta_{\hat{\theta}}(\theta)$, where $\hat{\theta}$ may be the ML or MAP estimate, then the posterior predictive density is obtained by simply plugging in the parameters, to yield a virtually identical rule:

$$p(y = c|x, \mathcal{D}) \propto \hat{\pi}_c \prod_{j=1}^{D} (\hat{\theta}_{jc})^{\mathbb{I}(x_j=1)} (1 - \hat{\theta}_{jc})^{\mathbb{I}(x_j=0)} \tag{14}$$

The only difference that the posterior mean θ is replaced with the posterior mode or MLE $\hat{\theta}$. However, this small difference can be important in practice, since the posterior mean will

result in less overfitting. The corresposing classiefier from (10) can also be simply written as follows

$$\text{classify}(f_1, \ldots, f_n) = arg \max_c p(C = c) \prod_{i=1}^{D} (F_i = f_i | C = c) \tag{15}$$

Although far-reaching independence assumptions are frequently incorrect, the decoupling of the class conditional feature distributions means that each distribution can be estimated independently as a one-dimensional distribution. This, in turn, aids in alleviating problems caused by the curse of dimensionality, such as the requirement for data sets that scale exponentially with the number of features. Like all probabilistic classifiers under the MAP decision rule, it arrives at the correct classification as long as the correct class is more probable than any other class; thus, class probabilities do not have to be estimated very well. [2]

III. Applications in Real-Life Problems

A. Data Description and Missing Values Handling

The dataset comprises information about households divided into two categories based on their annual income, i.e. the households who earn more than 50K and those who are below the 50k mark. Information about the educational background, relationship status, sex, race, native country, change in capital, the working class is given. The analysis aims to devise a generalised model that can determine whether a person earns more than 50K annually or not. Once the data is imported it is highlighted that there are missing values in the columns containing information about **workclass (5.6%)**, **occupation (5.7%)** and **native country (1.8%)**. From Fig. 1 it is observed that most people belong to the private working-class, works in an Exec-managerial role and belong to the United States. The number of missing values is significantly less. Therefore they had been imputed with the corresponding mode values.

B. Exploratory Data Analysis

The following section will focus on the relationship between different features and identify the most suited for modelling purposes. All the features would be used of modelling purposes. The number of people from each category who earn more than 50K is used a parameter to identify the relationship between income and given variables.

From Fig. 2 it can be conclude that the individuals with significant amount of education (like Bachelors, Masters, Doctorate, Prof-school) are more likely to fell in the category of earning more than 50k annually. On the other hand, from Fig. 3 it is observed both husband and wife are more likely to generate higher income in comparison to other groups.

It is observed that the number of high earning males is more than twice that of the number of females. In Fig. 4 the country-wise distribution of high income generating household is given. People from countries like India, Germany, France, England are more likely to earn $50K or more annually. In contrast, nearly all the nations have a higher number of males with a high income, which validates our earlier observation.

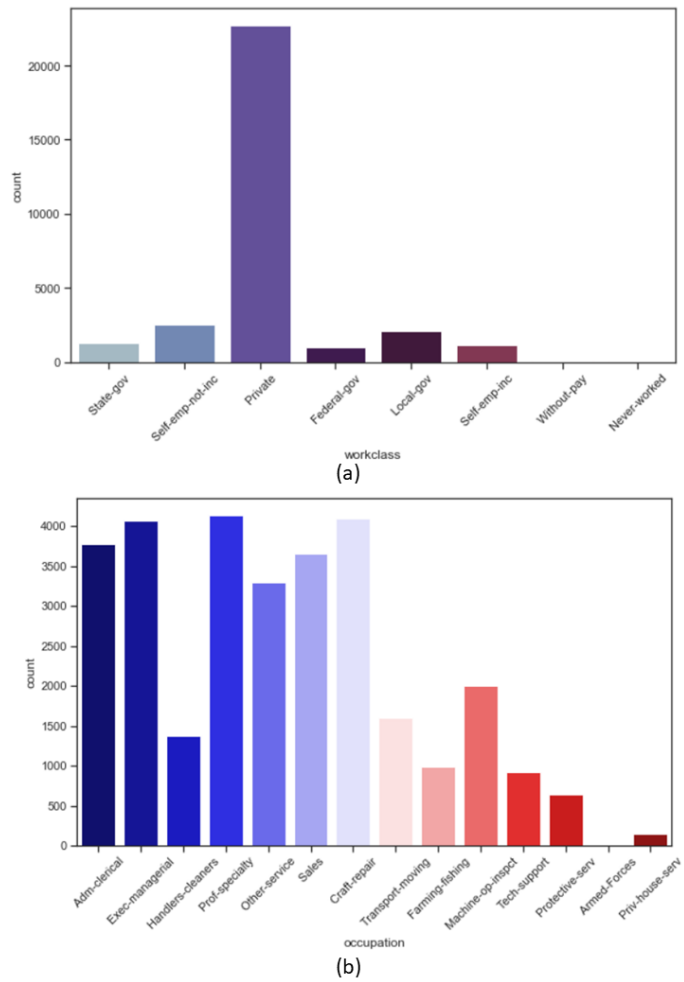

Fig. 1. (a) Frequency of individuals from different working class; (b) Frequency of individuals from different occupational backgrounds

On analysing the distribution of income concerning the working class of people, it is observed that a higher percentage of self-employed people are earning more than $50 K per year compared to other groups. It is against the fact that the most number of people works in the private sector. From Fig. 5 it can be inferred that black people are less likely to earn a higher income than people from other races despite being in the same working-class and have a very high unemployment rate even after being a minor part of the population.

The number of people generating higher income is directly proportional to the number of years spent on education as depicted by Fig. 6. This also validates the earlier observation that people who have achieved higher education are more likely to fall in the category of high-income earners. People who are relatively less educated are more likely to earn below $50K annual, which indicates that education is an essential factor in determining the yearly income of individuals.

The variable *fnlwgt* doesn't have a significant impact on the income status of people, and it is distributed evenly across the two given income groups. The distribution of martial_status with income indicates that households with two earning indi-

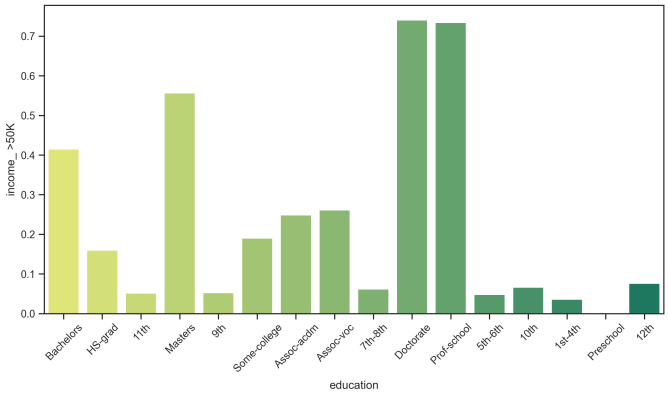

Fig. 2. People earning more than 50k based on education level

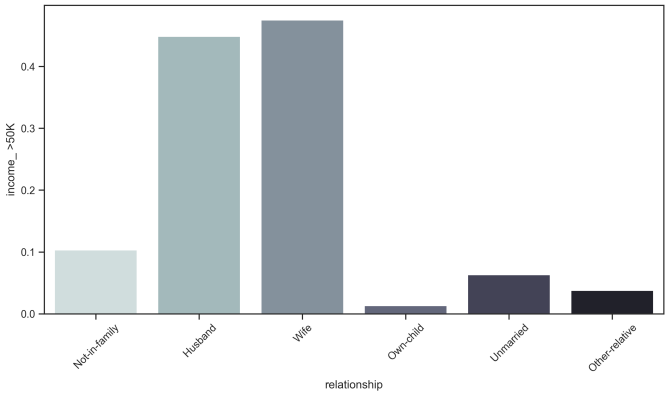

Fig. 3. Relationship vs high income distribution

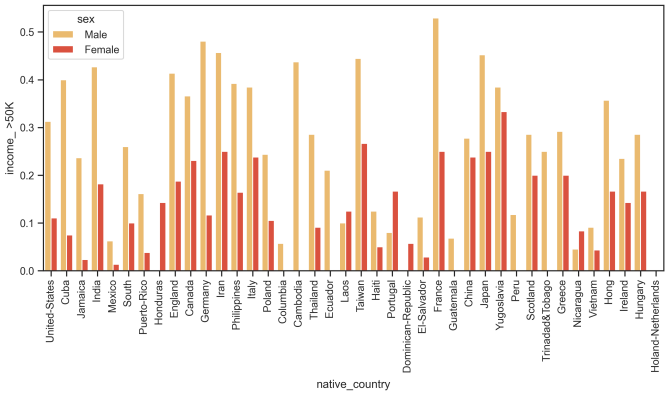

Fig. 4. Gender wise distribution of individuals earning more than $50K based on native countries

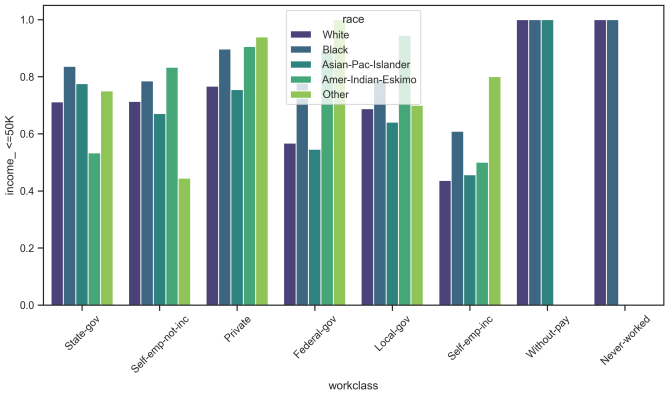

Fig. 5. Working class vs income less than $50k according to the race of individual

viduals (i.e. husband and wife) are more likely to fall in the category of high-income earners. [3]

Similarly it is also observed that on an average people with high earning are putting more hours per week for work in comparison to their counter parts. See Fig. 8.

Based on the observations above the following feature set is selected for modelling purposes:

- age
- workclass
- fnwlgt
- education
- education_num
- marital_status
- occupation
- relationship
- race
- sex
- capital_gain
- capital_loss
- hours

The correlation heat-map in the Fig. 9 doesn't indicate any strong correlation between the given variables, strengthening the assumption that these features are independent of each other and the chances of a naive Bayes classifier yielding good results are high.

C. Naive Bayes Classifier Model and Results

The following section will focus on the relationship between different features and identify the most suited for modelling purposes. The purpose of this paper is to identify the income group of a house-old and want to identify its relationship with the other features, the most crucial being age, education, race, native_country and sex. To do so, a generative model is fitted (which aims at learning the distribution of target classes rather than just identifying the decision boundary) is deployed. The mathematics behind the estimation of the line of best fit is described in detail in Sec. II. For our given problem our predictor is income and regressor consist of features describes in Sec. III-B.

The python package *sklearn* is used for performing the task of fitting a naive Bayes classifier. Scikit-learn (Sklearn) is one of the most valuable and robust libraries for machine learning in Python. It provides a selection of efficient tools for machine learning and statistical modelling, including classification, regression, clustering and dimensionality reduction via a consistent interface in Python. This library,

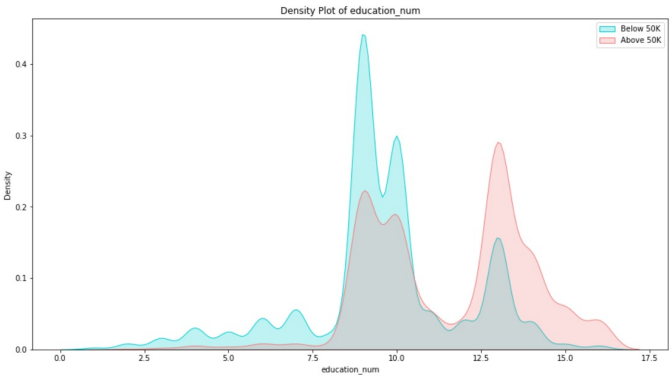

Fig. 6. Years spent on Education vs income

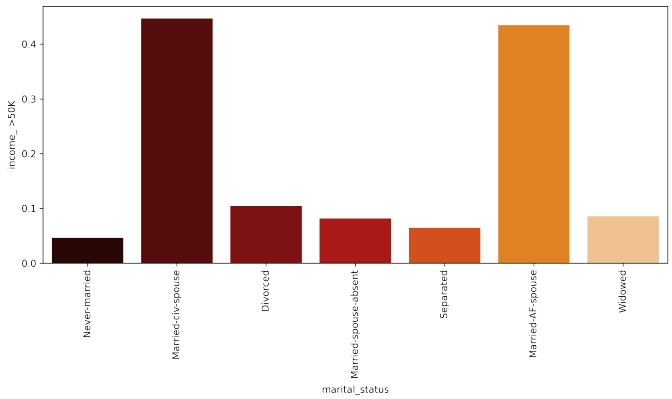

Fig. 7. Marial status vs income more than 50K

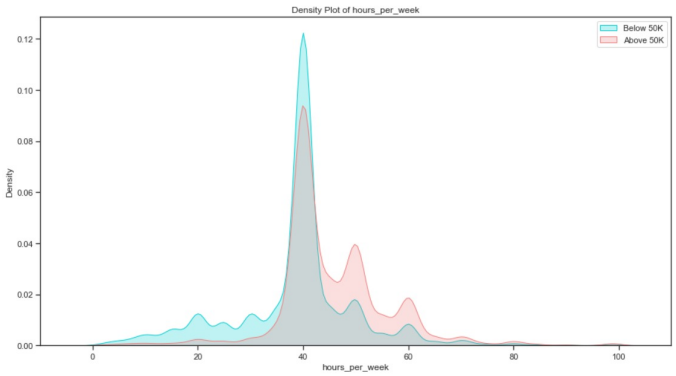

Fig. 8. Hours per week vs income

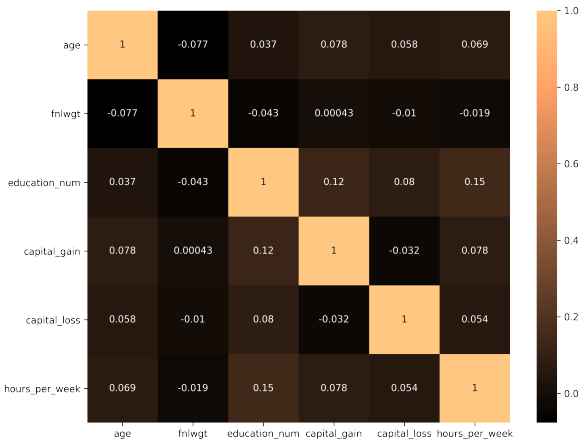

Fig. 9. Correlation heat-map

which is primarily written in Python, is built upon NumPy, SciPy and Matplotlib. The sklearn package provides different classes for the classification problem, including **GaussianNB**. Before proceeding with the modelling, one-hot encoding is performed on the non-numerical features like work-class, education, marital_status, occupation, relationship, race, sex and native_country. Robust scaler is used to scale the data. It subtracts the median value from data and then divides them by interquartile range. Therefore it is robust to outliers. Once the scaling is done the data is splited in train and test data. Afterwards a Gaussian naive Bayes model is fitted on the training data and the labels for test data is predicted.

The mean accuracy score on the test data obtained by the naive Bayes classifier is $\mathbf{0.8009}$, whereas that on the training dataset is $\mathbf{0.8007}$. The majority of households lies in the category of annual income less than \$50K. The **null accuracy** obtained is $\mathbf{0.7582}$. Using a **random guess strategy** indicates that using a probabilistic outcome of 0.8 will result in mean accuracy of $\mathbf{0.65}$, which further indicates that the classifier is doing a decent job. The confusion matrix looks as follows:

$$\begin{bmatrix} 5917 & 1907 \\ 1490 & 455 \end{bmatrix}$$

True Positive (TP): $\mathbf{5917}$, True Negative (TN): $\mathbf{1907}$, False Positive (FP): $\mathbf{1490}$, False Negative (FN): $\mathbf{455}$.

The classification report suggest the following

	precision	recall	f1-score	support
$<= 50K$	0.93	0.80	0.86	7407
$> 50K$	0.56	0.81	0.66	2362
accuracy			0.80	9769

Dropping feature **fnlwgt** did improve the accuracy on the test set but results in a drop in cross-validation accuracy, whereas lowering any other features resulted in a severe decrement in training and test accuracy itself. From our above observations, it can be inferred that the naive Bayes classifier gives decent results for identifying households where the annual income is more than \$50K. Black people, people from 3rd world countries and females are less likely to be in the higher income region than others.

IV. Conclusion

The study of the question: *what sorts of households are more probable to earn more than \$50K annually?* is essential to understand the bias while providing different private and

public policy benefits to people. If the education status of people is known, then those with higher qualifications are preferred in a job over others. Similarly, men's are more likely to earn higher salaries. From our observations, it can be reasoned that there is some kind of bias for people belonging to different races and women, which highlights a social issue and would come in handy for future policy-making. The EDA and naive Bayes classifier approach gives some decent idea about the relationship between predictors and regressors. It doesn't explain the underlying pattern thoroughly. The reason might be the presence of high cardinality in features like native_country and improper handling of parts. Future work will focus on measuring the impact of those features and performing feature engineering. A need to improve the model performance has also been identified and will be taken care of.

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